

# CS 3700 – Lab 03: Normalization and Crow's Foot ERD

Lastname, Firstname — Business Case: Clinic Management System (Lab 01)

## UNF

```
Appointment(PatientID, PatientName, DOB, Phone, Email, DoctorID, DoctorName,
Speciality, OfficeNumber, DepartmentID, DepartmentName, Location, AppointmentID,
AppointmentDate, AppointmentTime, Reason)
```

### Possible Candidate Keys

- PatientID
- DoctorID
- DepartmentID
- AppointmentID

AppointmentID uniquely identifies each row of the UNF table (one row per appointment); PatientID, DoctorID, and DepartmentID are the identifiers that will become primary keys of their own tables once the data is decomposed.

## 1NF

Phone is a multivalued attribute — a patient can have more than one phone number — so it is removed into its own relation to make every value atomic.

```
Appointment(PatientID, PatientName, DOB, Email, DoctorID, DoctorName, Speciality,
OfficeNumber, DepartmentID, DepartmentName, Location, AppointmentID,
AppointmentDate, AppointmentTime, Reason)
PatientPhone(PatientID, Phone)
```

## 2NF

2NF removes partial dependencies. PatientName, DOB, and Email depend only on PatientID; DoctorName, Speciality, OfficeNumber, and DepartmentID depend only on DoctorID; DepartmentName and Location depend only on DepartmentID. Each group is moved into its own table, leaving only AppointmentDate, AppointmentTime, and Reason (plus the PatientID/DoctorID foreign keys) in Appointment.

```
Patient(PatientID(PK), PatientName, DOB, Email)
Doctor(DoctorID(PK), DoctorName, Speciality, OfficeNumber, DepartmentID(FK))
Department(DepartmentID(PK), DepartmentName, Location)
Appointment(AppointmentID(PK), AppointmentDate, AppointmentTime, Reason,
PatientID(FK), DoctorID(FK))
PatientPhone(PatientID(PK,FK), Phone(PK))
```

## 3NF

3NF removes transitive dependencies — non-key attributes that depend on another non-key attribute instead of directly on the primary key. Checking each 2NF table: in Doctor, DepartmentID determines DepartmentName and Location, but those attributes were already moved out into Department in the 2NF step (Doctor only keeps the DepartmentID foreign key), so no transitive dependency remains. No other non-key attribute in any table depends on another non-key attribute. The schema is therefore already in 3NF with no further decomposition needed.

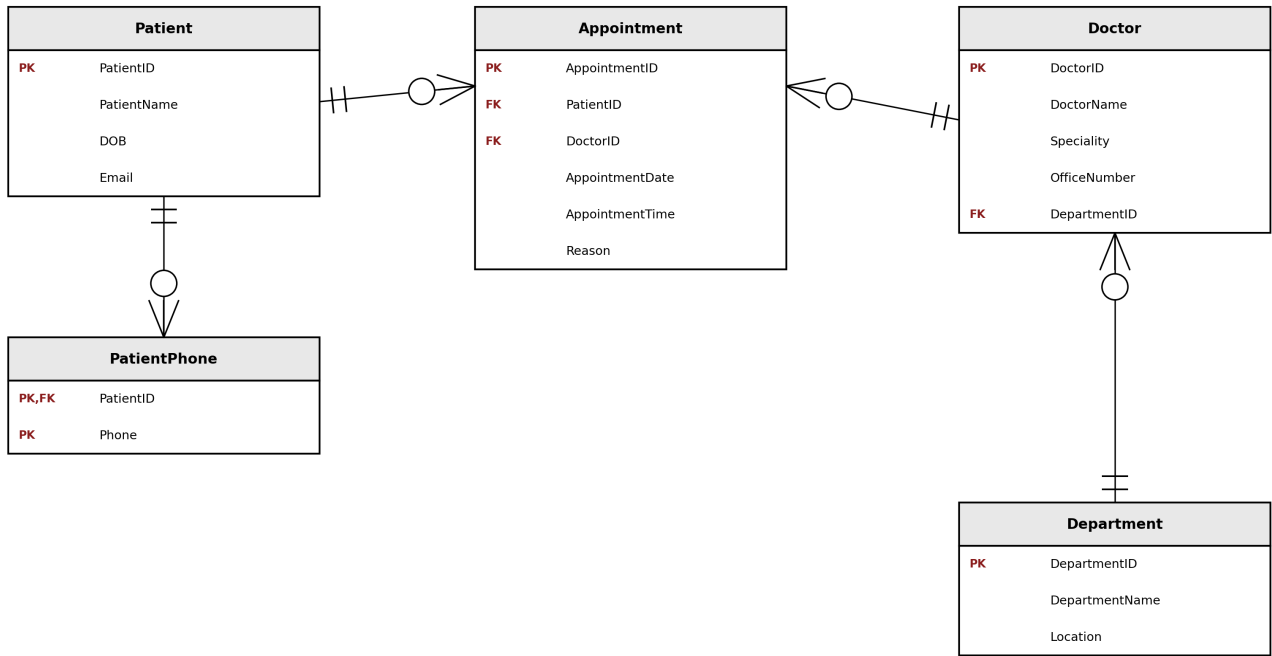
### Final 3NF Relations

| Patient        | PatientPhone      | Doctor            | Department        | Appointment        |
|----------------|-------------------|-------------------|-------------------|--------------------|
| PatientID (PK) | PatientID (PK,FK) | DoctorID (PK)     | DepartmentID (PK) | AppointmentID (PK) |
| PatientName    | Phone (PK)        | DoctorName        | DepartmentName    | AppointmentDate    |
| DOB            |                   | Speciality        | Location          | AppointmentTime    |
| Email          |                   | OfficeNumber      |                   | Reason             |
|                |                   | DepartmentID (FK) |                   | PatientID (FK)     |
|                |                   |                   |                   | DoctorID (FK)      |

```
Patient(PatientID(PK), PatientName, DOB, Email)
PatientPhone(PatientID(PK,FK), Phone(PK))
Doctor(DoctorID(PK), DoctorName, Speciality, OfficeNumber, DepartmentID(FK))
Department(DepartmentID(PK), DepartmentName, Location)
Appointment(AppointmentID(PK), AppointmentDate, AppointmentTime, Reason,
PatientID(FK), DoctorID(FK))
```

# Crow's Foot ERD

Clinic Management System — Crow's Foot ERD (3NF)



**Legend:**  
—||— one and only one    —o< zero or many  
PK = Primary Key    FK = Foreign Key